

H

 LV.0 → LV.1 · AI ENGINEER

The 6 H's that move you from **Level 0** → **1** AI engineer.

Hallucination, Hybrid Search, Hyperparameters and more. **Plain-English** definitions, one swipe each, plus the platforms the whole field builds on.

HYPOTHESIS TESTING • FOUNDATIONS

Hypothesis Testing

claim • vs • chance

Is this result real or just luck? Hypothesis testing weighs the evidence against a boring default explanation and tells you whether you can **reject it**. It is the quiet engine inside every A/B test.

USED IN

- A/B testing
- Experiments
- Statistics

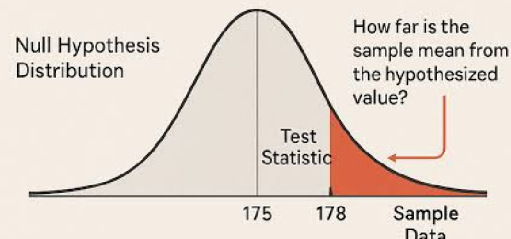
Understanding Test Statistics: A Comprehensive Guide

What is a Hypothesis Test

Null Hypothesis (H_0)
Population mean = 175

Alternative Hypothesis
Population mean $\neq$ 175

The Role of the Test Statistic



$$\text{Test Statistic} = \frac{\text{Sample Statistic} - \text{Population}}{\text{Standard Error}}$$

$$t = \frac{178 - 175}{s/\sqrt{n}} \quad t = 3.29$$

Interpreting the Test Statistic

- Compare to a Critical Value
- Calculate a p-value

NULL VS ALTERNATIVE

Hyperparameters

set • not • learned

Some settings the model learns. These it does not. Learning rate, batch size, layers, and epochs are **knobs you set before training starts**, and dialing them in often separates a model that generalizes from one that flops.

USED IN

- Model training
- Grid / Bayesian search
- AutoML

learning rate • $3e-4$

batch size • 32

layers • 12

epochs • 5



MODEL
trains & learns

KNOBS YOU SET

HALLUCINATION • ALIGNMENT

Hallucination

confident • but • wrong

Confident, fluent, and completely wrong. A hallucination is what you get when a model **predicts likely words instead of checking facts**. Grounding, retrieval, and tool use are how we keep it honest.

USED IN

- LLM reliability
- RAG
- Evaluation

8 steps for preventing AI hallucinations



PREVENTION STEPS

HYBRID SEARCH • RETRIEVAL

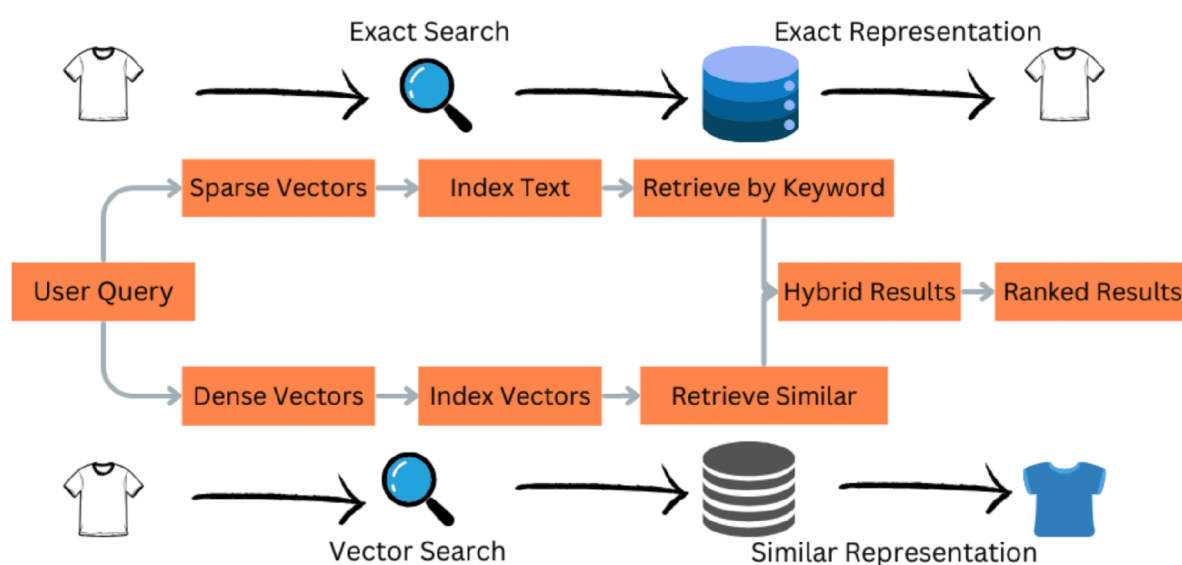
Hybrid Search

keyword + meaning

Keywords catch the exact words; vectors catch the meaning. Hybrid search runs **both and merges the results**, which is why modern RAG retrieves better than either approach manages alone.

USED IN

- RAG retrieval
- Vector DBs
- Search ranking



KEYWORD + VECTOR → RANKED

HUMAN-IN-THE-LOOP • AGENTS

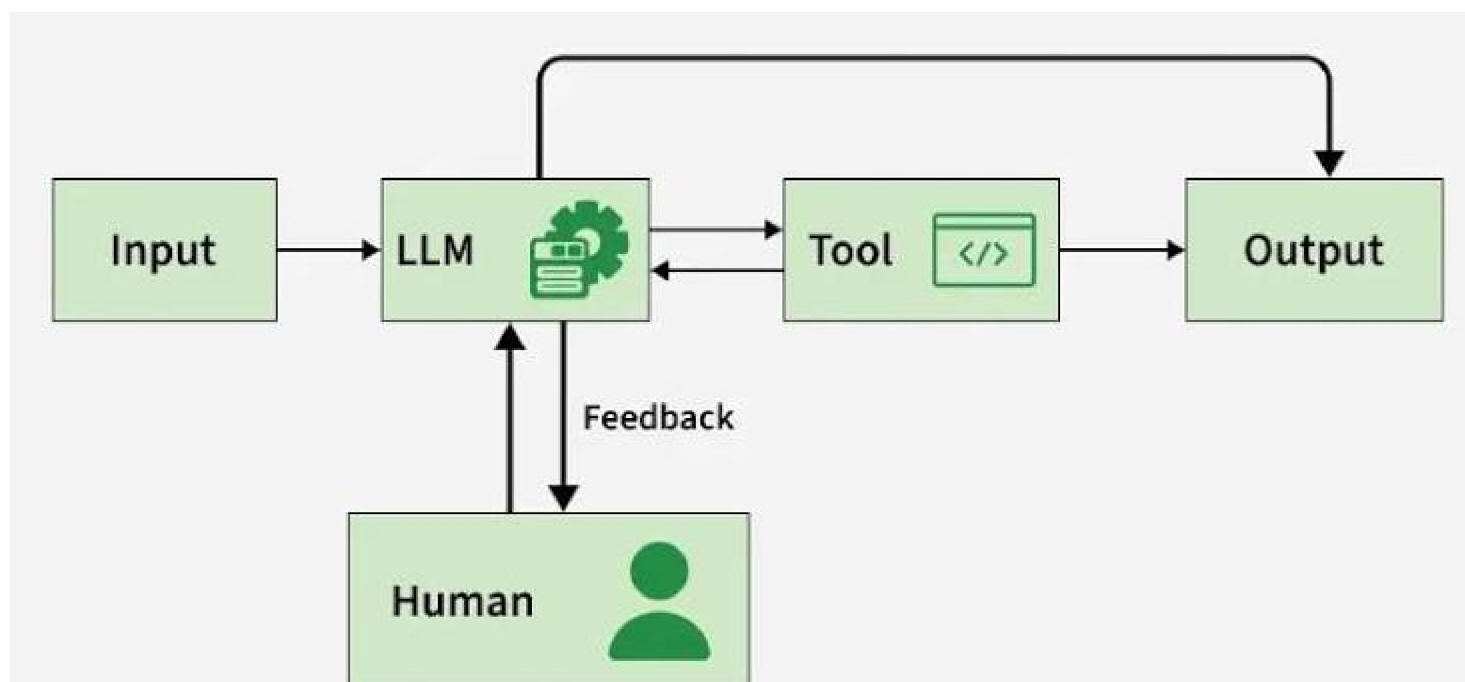
HITL

AI • proposes • human approves

Keep a human in the loop and you get machine speed with human judgment. HITL has a person **review, correct, or approve AI output**, essential anywhere accuracy and accountability truly matter.

USED IN

- Agent oversight
- RLHF
- High-stakes AI



HUMAN FEEDBACK LOOP

HEARTBEAT • INFRASTRUCTURE

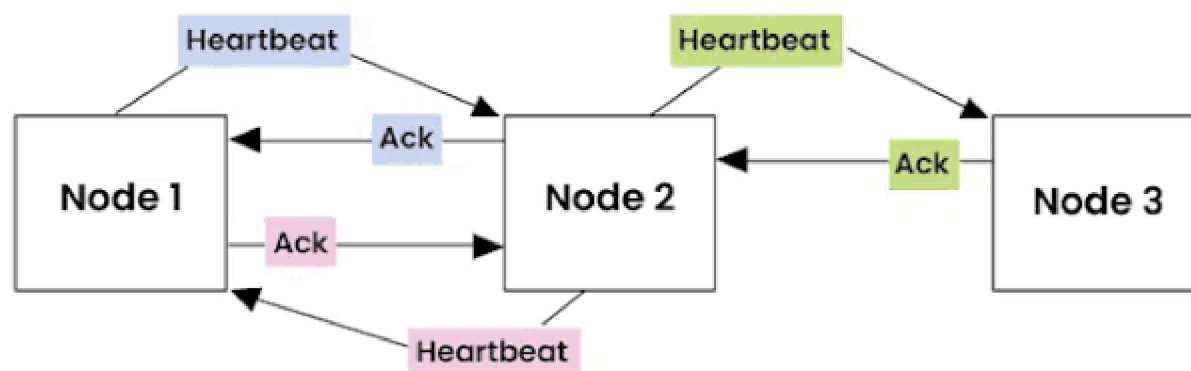
Heartbeat

"still alive" • signal

A steady pulse that says "still here." Nodes send heartbeats so the system knows they are alive; **miss a few and the others reroute**, re-elect, or restart. It is the basis of failure detection.

USED IN

- Distributed systems
- Health checks
- Failover



What are Heartbeat Messages?



NODES PING EACH OTHER

↓ UP NEXT • TOOL

Tool next.

Beyond the **6 H-words**, the platform that hosts the models, datasets, and demos the whole field runs on.



OPEN-SOURCE AI PLATFORM

Hugging Face

models • datasets • spaces



Hugging Face

Think GitHub, but for AI.
Hugging Face hosts **millions of open models, datasets, and demos**, plus the Transformers library an awful lot of the field is quietly built on.

WHAT IT IS

An open hub for models, datasets & demos.

IN AI STACKS

Transformers, Datasets, Inference, Spaces.

WHY DEVS USE IT

Pull, fine-tune, and ship models fast.

PAIRS WITH

PyTorch, LangChain, your own pipelines.

DISTRIBUTED DATA WAREHOUSE

Hive

SQL • over • Hadoop at scale



SQL, meet big data. Apache Hive lets analysts **query enormous datasets on Hadoop** using familiar syntax, hiding all the distributed-computing machinery underneath.

WHAT IT IS

SQL-on-Hadoop data warehouse system.

IN AI STACKS

Feature stores, ETL, ML data prep.

WHY TEAMS USE IT

Query petabyte-scale data with SQL.

PAIRS WITH

Hadoop, Spark, Presto, dbt.

AI VIDEO GENERATION

Higgsfield

prompt • → • cinematic video



Type a scene, get a film.
Higgsfield turns prompts into
cinematic, social-ready clips,
with camera moves and effects
that make AI video look
directed rather than generated.

WHAT IT IS

AI video generator for short
cinematic clips.

IN AI STACKS

Marketing, social, creative
pipelines.

WHY CREATORS USE IT

Camera moves & VFX from a
prompt.

PAIRS WITH

Sora, Runway, Veo, image gens.

↓ UP NEXT • PEOPLE TO FOLLOW

Spotlight next.

Two **high-signal, low-noise** educators worth a follow to keep your H-stack current.

H

HIGH-SIGNAL, LOW-NOISE

Hitesh Choudhary

X • @Hiteshdotcom • educator



One of India's most-followed coding teachers, with a build-first philosophy. From web dev to LLMs and agents, **his lessons stick because you ship something** by the end of them.

WHAT THEY POST

Hands-on coding, AI, and career content.

FIND THEM

X @Hiteshdotcom, YouTube & Instagram.

WHY FOLLOW

Build-first lessons that actually stick.

BEST FOR

Going from concept to shipped project.

HIGH-SIGNAL, LOW-NOISE

Harsh Bhatt

X • @harshbhatt7585 • AI engineer



Coding since he was twelve, Harsh makes sharp explainers on open-source models and building AI from scratch. A rare mix of **real depth and the gift for making it click.**

WHAT THEY POST

Open-source models, built-from-scratch AI.

FIND THEM

X @harshbhatt7585, LinkedIn.

WHY FOLLOW

Deep expertise, simply explained.

BEST FOR

Understanding models from first principles.

The H stack, in one card.

Save it for later. Drop a comment if there's an H-term you'd add.

01 **Hypothesis Testing** foundations

02 **Hyperparameters** learning

03 **Hallucination** alignment

04 **Hybrid Search** retrieval

05 **HITL** agents

06 **Heartbeat** infra

+ **Hugging Face** tool

+ **Hive** tool

+ **Higgsfield** tool

@ **Hitesh Choudhary** people

@ **Harsh Bhatt** people



FOLLOW FOR THE FULL A TO Z

Prasanna Kulkarni

in/prasanna-kulkarni-032650180

| Next